
First Line of the Title

Second Line of the Title

Nuclear Engineering – Politecnico di Milano

Pagliuca Simone, et al.

my contact, al.'s contacts

Course: Or other indication

Academic Year: Or some time indication

ABSTRACT

Here write you Abstract.

Key-words: Key, Words, Here

NOMENCLATURE

Type A

x X quantity

Type B & Misc

y Y quantity

CONTENTS

1 INTRODUCTION

1.1 Context

We are in 2050 Temporale, geografico

1.2 Case Study

The study will simulate and SMR installed in a prevalently residential area. The reactor will be coupled with a secondary system and the electricity production should be able to cover the entire demand of the area. The model also takes into account the share of renewable energy sources in the area.

1.2.1 Reactor

Our plant consists in an SMR installed in the Joint Research Center (JRC) in Ispra, Northern Italy. The JRC is a research facility of the European Commission, located on the shores of Lake Maggiore. The facility is home to the ESSOR reactor, currently facing decommissioning. The facility has been chosen since it represents an already nuclearized site which suggests the possibility of a future nuclear plant. Moreover the installation of the SMR could take place, technically, withing the ESSOR reactor considering that the containment building was oversized due to the particular use case of the ESSOR reactor in order to accomodate not only the reactor itself but also several laboratories.

1.2.2 Secondary Plant

1.3 Objectives

The objective of this work is to determine the feasibility of this installation both technically and economically. On the technical side of things we focus on the capability of the system to cover the energy demand and to adapt to transient conditions. The economical side of things will assess the minimum total plant cost that would allow for a positive NPV. We are going to evaluate 4 scenarios:

- Today: the current state of the residential sector in the area
- Optimistic Scenario: the residential sector is mostly electrified with highly efficient homes
- Pessimistic Scenario: the residential sector is mostly electrified with low efficiency homes
- Future Scenario: the residential sector is mostly electrified with highly efficient homes and a

The scenarios are translated as a distribution of energy classes in the residential sector: IMMAGINI DELLE 4 DISTRIBUZIONI.

We are going to evaluate the effectiveness of the hybrid system by means of the following key parameters:

- The investment cost to make the plant profitable
- The decarbonization impact DOPO SPIEGHI MEGLIO
- Transient response of the system with an hourly resolution

1.3.1 Decarbonization Impact

E una roba che ci inventiamo noi spiga meglio :D

2 DATA COLLECTION

At the basis of our work is the knowledge of a realistic energy demand profile, that would catch the transients throughout the year. Given our focus on the residential sector, the overall energy demand can be seen as the sum of the heating/cooling contribution + all the other electricity consumptions for appliances and lightning. Several simplifications will be made through the following derivation but we point out any possible improvement that could be made but was considered out of the scope of this work.

2.1 Approach

2.1.1 Introduction

In Italy, an APE (Attestato di Prestazione Energetica) is a document that certifies the energy performance of a propriety unit (unità immobiliare). This certification must be issued by a qualified technician and is mandatory for all buildings that are sold, rented or refurbished.

The APE provides information about the energy consumption of the property, its energy class. The energy class is a letter grade (from A4 to G) that indicates the energy performance of the building, with A4 being the most efficient and G being the least efficient.

Each unit refers to a distinct and identifiable part of a real estate property, such as an apartment, a house, or a commercial space, that can be individually owned, sold, or rented. It is a specific segment of a larger property that has its own legal and functional identity. For instance, a building with three apartments will have three separate APEs, one for each apartment. For this reason we can safely assume that each unit is the residence of a single family.

2.1.2 Methodology

We plan to determine the demand profile of electricity in the region $P_{e,tot}$.

From simulations we are able to retrieve the hourly demand profile of thermal energy for heating $\dot{Q}_h$ and cooling $\dot{Q}_c$ for each energy class i . We then evaluated a reference-day profile for the thermal demand for domestic hot water $\dot{Q}_{dhw,t}$ which is assumed to be constant throughout the year, based on REFERENCE. This is converted to electicty demand by taking into account the share of buildings in that energy class that use a heat pump for climatization purposes χ_{HP} and its efficiency for heating COP and for cooling EER . To this we then add a contribution of electric consumption for other purposes P_{other} namely lighting and appliances.

$$P_{e,i}^- = \left(\frac{\dot{Q}_{h,i}}{COP} + \frac{\dot{Q}_{c,i}}{EER} \right) \cdot \chi_{HP} + P_{other} \quad (1)$$

ESPANDI SULLA QUESTIONE FOTOVOLTAICO Then we estimate the production of photovoltaic systems for each energy class $P_{e,i}^+$. Finally the reference energy demand for each energy class can be obtained by subtraction of demand and production: $P_{e,ref,i} = P_{e,i}^- - P_{e,i}^+$

Then, we can simply sum the contributions, each wheighted on their respective class share χ_i , to get a reference profile of the region under study. AGGIUNGI REFERENCE ISTAT E CAZZI This can be multiplied by the number of buildings in the region N_{UI} to obtain the total demand $P_{e,tot}$.

$$P_{e,tot} = N_{UI} \cdot \sum_{i=A4}^G P_{e,ref,i} \cdot \chi_i \quad (2)$$

2.1.3 Data Sources

From the CENED database [**cened2025**] we extracted the information about the relevant municipalities of the region, which we derived based on the map of primary substations from the national grid operator (GSE) [**gse_cabine_primarie**]. The data was also filtered to include only residential buildings, excluding commercial and industrial ones. This was necessary given the large amount of data from the complete regional database. This way an easier to manage csv file was obtained. AGGIUNGI CHE HO TOLO VALORI WEIRD Note that an API is available to access the database but it was found to be very slow and inefficient.

2.2 Heating and Cooling Demand

For each energy class we have taken a real example of a residential building from the region under study and computed the hourly thermal demand for heating and cooling throughout the year. This is obtained by means of a professional software (TERMOLOG) that allows for dynamic simulation of the energy system of the building. This computation is based on the UNI EN ISO 52016 standard. We used one reference building per energy class all with the same utilization profile, which is a simplification to be discussed later. The output of the calculation is an hourly profile of the thermal demand for heating and cooling.

2.2.1 Normalization of the Data

RISCRIVI E TUTTO SBAGLIATO

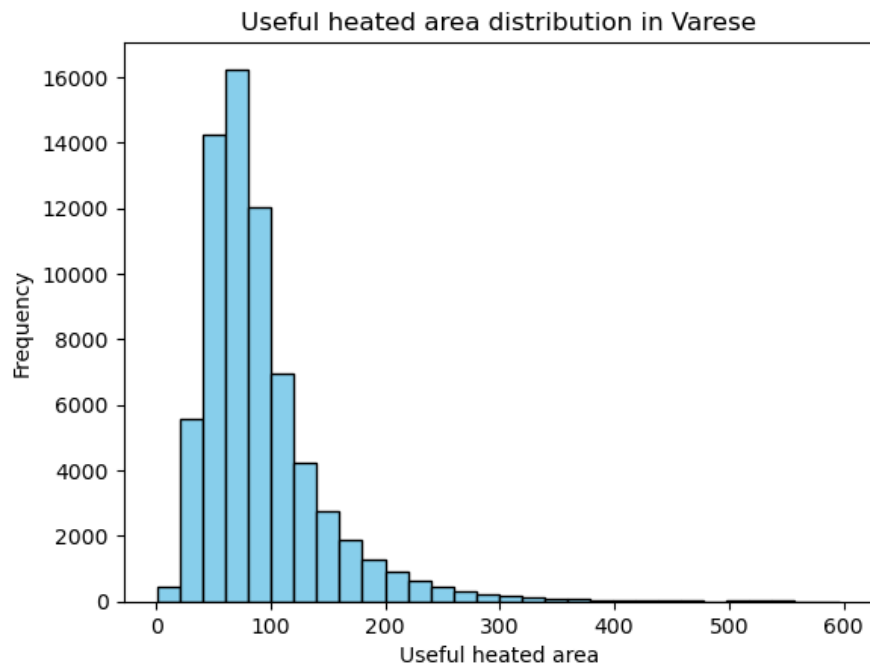


Figure 1: Histogram of the useful heated area of buildings.

2.3 By class distributions

To be able to simply simulate different scenarios we have identified characteristics of each energy class so that we can just modify the energy class distribution and all the parameters needed to determine the energy demand will follow.

2.3.1 Heat Pump distribution and Cooling Demand

We identified the share of heat pumps per energy class. We also determined the share of buildings that have a cooling system installed, which as of today comes up to 9.58%. This is a very low share and is to be expected as the region under investigation is quite chill during summer.

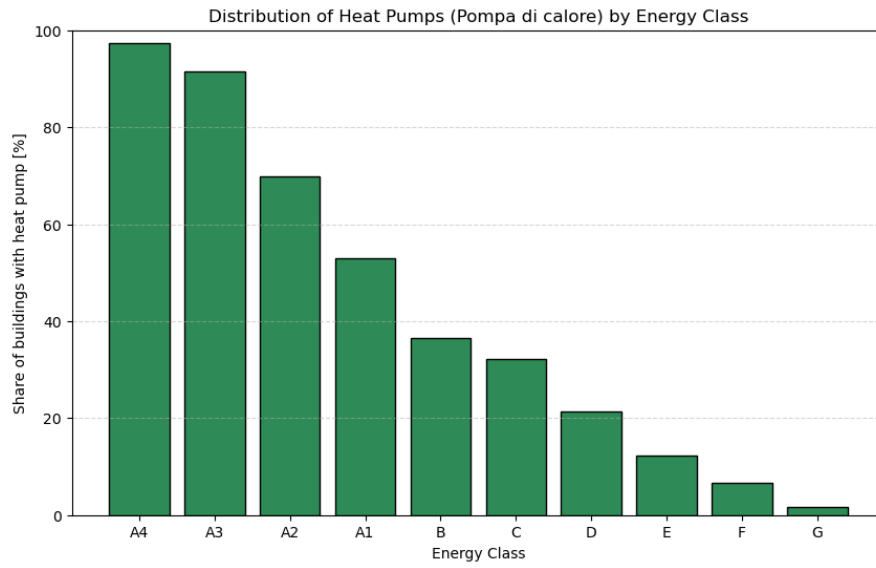


Figure 2: Distribution of heat pumps across energy classes.

2.3.2 Photovoltaic Distribution

We identified the share of building that have photovoltaic systems installed per energy class.

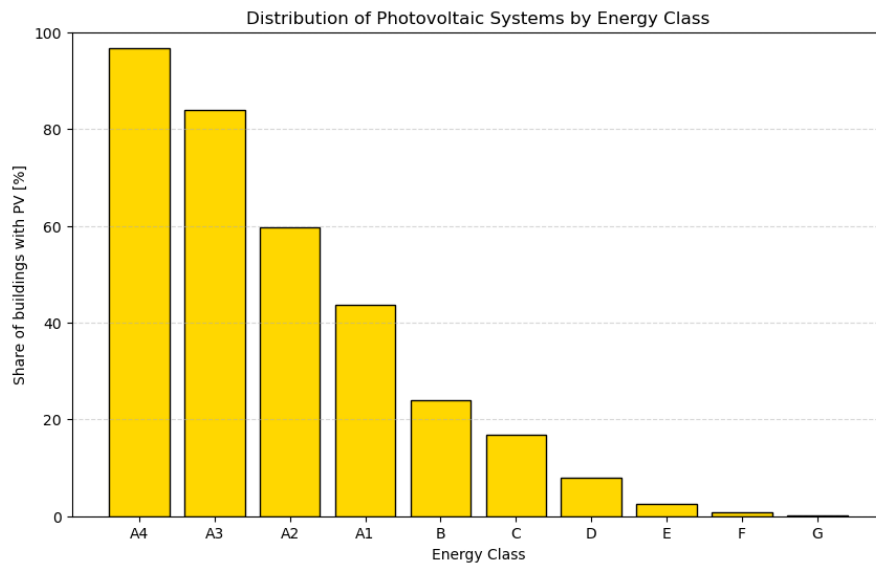
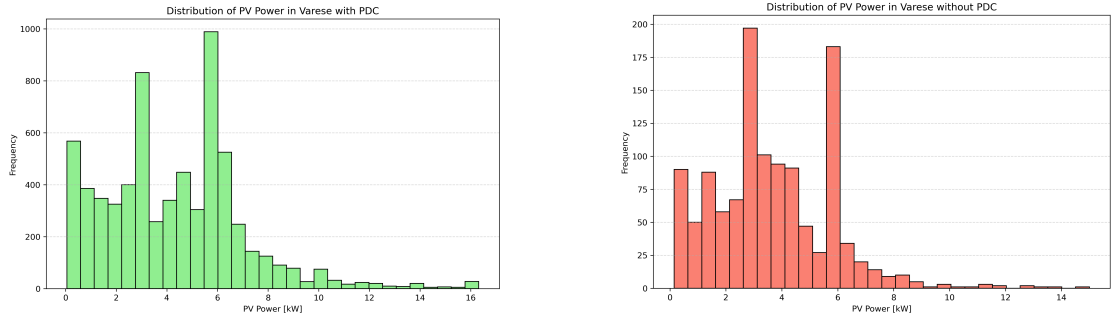


Figure 3: Distribution of photovoltaic systems across energy classes.

Moreover we then divided the data into two groups, those with an heat pump and those without. From these we plotted the distribution of the size of the photovoltaic system installed. And computed the average (of the P_{99})



(a) Photovoltaic system size distribution for buildings with heat pumps.

(b) Photovoltaic system size distribution for buildings without heat pumps.

Figure 4: Distribution of photovoltaic system sizes based on heat pump presence.

The averages values are: 4.30kW and 3.70kW for the sistem with and without heat pumps respectively.

Observation on the relative sizes of PV systems VERIFICA SE HA ANCORA SENSO DI ESISTERE
Another interesting information we gathered is the relative size of the photovoltaic system. Using data from the CENED database: Exported Electricity, Imported Electricity and In situ consumption.

$$\text{SCF: Self-Consumption Factor} = \frac{\text{In situ consumption}}{\text{Import} + \text{In situ consumption}} \quad (3)$$

$$\text{OSF: Over-Sizing Factor} = \frac{\text{Export} - \text{Import}}{\text{Import} + \text{In situ consumption}} \quad (4)$$

Whie the SCF is quite obvius, we defined the OSF to consider the resonable oversizing of the system that ensures that the energy exoported covers, at least, the imported energy. A value of 0% indicates a balanced system, positive values indicate greater export then import, negative the opposite.

Class	SCF (%)	OSF (%)
A4	38.49	65.47
A3	55.92	78.28
A2	68.32	299.50
A1	74.43	1029.90
B	80.71	966.47
C	84.49	434.58
D	90.84	111.82
E	96.08	120.58
F	98.85	-67.53
G	99.76	-95.48

Table 1: SCF and OSF indicators by Energy Class.

The extreme overdimensioning on some classes can be explained as the result of two effects:

- Units with PV installed but using gas for heating.
- Simplified calculation of the CENED engine
- Counter effects of incentives and tax deductions.

Units in the middle of the energy class scale (A1 to D) are likely to still use gas for heating, as seen from the previous Figure ???. None the less a some of them still have a photovoltaic system installed. This means that most of the energy produced is not used on site.

Regarding the CENED engine, it is important to note that it uses a simplified calculation: it computes monthly averages, doesn't take into account energy storage, nor it consider any other time of electric consumption (such as appliances or lighting).

At last, it's known from professional experience that to boost the energy class, it is common that contractors choose to an over sized photovoltaic system. This practice was particularly common during the "superbonus" (2020-2023) period which required a mandatory jump of 2 energy classes to be eligible for the tax deduction. It could be interesting to evaluate the impact of this on the efficiency of the grid.

2.3.3 Electricity Production

We have to obtain a reference value of the photovoltaic production profile. First, the list of municipalities is fed to the Nominatim API [**Nominatim**] to obtain the coordinates of the center of each municipality. Then, we use the PVGIS API [**PVGIS**] to obtain the hourly production profile for each municipality. We fixed the angle to an average 22° and a plant size of $1kW$. For each municipality we obtain a profile every 10° of the azimuth angle, from 0° to 360° . Those are then weighted to a gaussian distribution to represent that it is preferred to point the photovoltaic system to the south. This gives us a reference photovoltaic production profile for each municipality $P_{e,m}^+$, where m is the municipality. Then we can obtain the reference production profile for each energy class $P_{e,ref}^+$ by weighting the production profile of each municipality on the share of buildings in that municipality over the total.

Finally to obtain the reference production profile we used the following formula:

$$P_e^+ = \sum_{i=A4}^G P_{e,ref}^+ \cdot (4.30\chi_{hp}(i) + 3.70(1 - \chi_{hp}(i))) \cdot N_b(i)\chi_{pv}(i) \quad (5)$$

2.4 Conversion from thermal to primary energy demand

UN PO UNA RIPETIZIONE VEDI SE TOGLIERE PRIMA E ESPANDERE QUESTA The dynamic simulation outputs the thermal demand of the building. To estimate the primary energy needed we used average efficiencies of the heating and cooling systems.

$$\dot{Q}_{fossil} = \frac{\dot{Q}_{th}}{\eta_{fossil}} \quad (6)$$

$$P_e = \frac{P_{th, heating}}{COP_h} + \frac{P_{th, cooling}}{COP_c} \quad (7)$$

2.5 Other Electricity Demand

Electrical demand for appliances and lighting is modeled equal for all classes. The definition was manual, hour by hour for an average day in the following "seasons":

- Winter Weekday
- Winter Weekend
- Summer Weekday
- Summer Weekend
- Winter Holiday
- Summer Holiday

Moreover we added an absence factor to take into account when people go on holiday massively during august.

The profile is defined by considering the following contributions:

- standby and always on appliances
- lighting
- appliances

During the summer the need for a cooling fan was added. Moreover we added a parameter to represent the share of buildings with an induction system installed.

2.6 Domestic Hot Water

QUI MANCA IL DHW !!!!!

2.7 Total Grid Demand

In the end we choose an energy class distribution to represent a future scenario. Given that we know the reference demand-production profile of each building we can sum them up by weighting on their supposed share to obtain the total reference demand which is then multiplied by the number of buildings in the region to obtain the total demand.

2.7.1 Total Buildings in the Region

2.7.2 Comparison and Tuning

To verify our method is acceptable we compare the total demand by the data available online, in particular we compared it to the average annual demand, in the residential sector for the province of Varese [[portaleconsumi_energia_domestica](#)], which happens to be 1931kWh in 2023, with a decreasing trend in the last years (2016kWh in 2022 and 2119kWh in 2021).

2.7.3 Demand Distribution

IMMAGINE DELLA DISTRIBUZIONE DELLA DOMANDA

3 MODELICA PLANT MODELLING

3.1 Plant Components

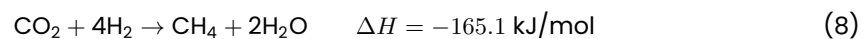
PRIMA DI TUTTO IL LAYOUT

3.1.1 SMR

3.1.2 H2 Plant

3.1.3 Methanation Reactor Modeling Approach

The methanation reactor was modeled in Modelica to simulate the conversion of hydrogen (H_2) and carbon dioxide (CO_2) into methane (CH_4) and water (H_2O) via the Sabatier reaction:



To approximate the level of CH_4 produced from a given feed, a dimensionless parameter `conversionEfficiency` was introduced. This factor (default: 0.78) reflects the maximum chemical efficiency of catalytic methanation as reported in Götz et al. [Götz2016]. It represents the fraction of the limiting reactant (either CO_2 or H_2) that is effectively converted to CH_4 , accounting for reactor inefficiencies and incomplete conversion at realistic temperature and pressure conditions.

The molar output flows are then calculated via stoichiometry:

$$\begin{aligned}\dot{n}_{CH_4} &= \dot{n}_{lim} \cdot \text{conversionEfficiency} \\ \dot{n}_{H_2O} &= 2 \cdot \dot{n}_{CH_4} \\ \dot{n}_{H_2} &= \dot{n}_{H_2,in} - 4 \cdot \dot{n}_{CH_4} \\ \dot{n}_{CO_2} &= \dot{n}_{CO_2,in} - \dot{n}_{CH_4}\end{aligned}$$

3.1.4 Turbine

3.1.5 Other BOP Components

Control room, buffers etc etc

3.2 Demand Definition

To run the model we decided to compute a priori the daily setpoint for:

- Turbine load

- HTSE load
- Steam tap flow

to do so .. SPIEGA PROCEDIMENTO DA INPUT FOR MODELICA SECTION DI ENERGY D_{EMAND}

3.2.1 Dati che diamo in input

3.2.2 Rate limiting

3.2.3 Clusters

spiega che fanno i clusters + cita che usiamo dei profili di alcuni giorni particolari per vedere risposta del sistema

4 ECONOMIC STUDY